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Course: ITAI-1378-Comp Vision-Artificial Intel

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Lab 04 - Chihuahua or Muffin Workshop

Executive Summary

This reflective journal evaluates the implementation and performance of a basic linear neural network engineered to try and solve the "Chihuahua vs. Muffin" challenge. Using the PyTorch framework, this study analyzes how a multi-layer linear network struggles to build clear decision boundaries over images with highly similar color profiles and textures. The report details the process of identifying template code bugs, configuring standard image transformation inputs, and analyzing why the default baseline architecture yields a random 50% guessing accuracy on fine-grained visual data.

1. Workshop Objectives and Technical Methodology

The main goal of this workshop is to train a machine learning model to separate Chihuahuas from muffins. The project uses a clear four-step process:

- **Data Prep:** We load the images into Google Colab. We resize every image to a standard 224x224 pixels. This size is required because the network layers are fixed. We also change the pixel values to a steady mathematical range between -1.0 and 1.0 to keep the training smooth.
- **Model Setup:** We build a model named MySkynet using basic linear layers (nn.Linear). The model flattens the image into one long row of numbers and passes them through three small hidden layers.
- **Optimization:** We will use Cross-Entropy Loss to measure how many mistakes the model makes. An optimizer updates the weights to fix those mistakes.
- **Testing:** We split our 150 images into an 80/20 split (120 images for training and 30 images for validation). We use a visualization grid to check the accuracy percentage for each test image.

2. Core Machine Learning Concepts Learned

- **Fine-Grained Classification:** This means separating objects that look almost identical. Chihuahuas and muffins share the same brown colors and dark spots (eyes vs. blueberries). This makes it very hard for a basic computer program to tell them apart.

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- **Linear vs. CNN Layers:** We learned that flat linear layers are not good for images. Flattening an image destroys the 2D shapes, like how eyes connect to a nose. True Convolutional Neural Networks (CNNs) are much better because they keep the 2D shapes intact.

3. Engineering Challenges and Solutions

- **The Final Layer Bug:** The original code had a major logic error. It used an `F.relu()` function on the very last output layer. ReLU turns all negative numbers into zero. This stopped the network from expressing confidence *against* a class, which broke the math. We fixed this by removing the `F.relu()` from the last layer so the raw outputs could flow cleanly into the final percentage calculator.
- **Stuck at 50% Accuracy:** The default settings used a learning rate of 0.1, which was way too high. The model made updates that were too aggressive and just guessed randomly (50% accuracy). We fixed this by switching the optimizer to Adam and lowering the learning rate to 0.001 (Experiment 1). This allowed the model to learn smoothly.

4. Analytical Insights and Empirical Extensions

Data Split Dynamics across Different Scales

An important insight from this workshop centers on how data volumes affect split ratios. This lab relies on a classic 80/20 data split (120 training images and 30 validation images). For small datasets, this distribution is necessary to ensure the validation pool is large enough to yield a statistically reliable evaluation score.

However, as data volumes scale into large industrial repositories (such as millions of e-commerce product photos), the allocation shifts to a **95/5** or **99/1** split. With millions of samples available, a small 1% slice still provides tens of thousands of unseen images for evaluation, allowing engineers to maximize the data dedicated to the training pipeline.

5. Potential Real-World Applications

- **Automated Medical Diagnostics:** The exact fine-grained classification models optimized in this lab translate directly to healthcare imaging. They can be used to distinguish between malignant carcinomas and benign dermal lesions, or to classify complex patterns in radiology data.

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- **Industrial Automated Quality Control:** These techniques can be deployed on high-speed assembly lines to catch micro-defects in manufacturing components, using shape-invariant feature mapping to flag structural irregularities.
- **E-Commerce and Digital Asset Management:** Large digital storefronts use these algorithms to automatically clean, tag, and organize millions of user-submitted product photos, ensuring accurate inventory classification without relying on manual entry.

6. Personal Reflection

This lab taught me that building deep learning models is about trial and error. Simply adding more layers does not fix a model if your code has bugs or if your settings are wrong. Fixing the template bugs showed me how important optimization is. This links directly to my previous e-commerce study: true engineering success comes from careful, data-driven optimization.

References

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